

Beyond Fixed Margins: Generalizing Adaptive Sparse Softmax

Sandesh Thakuri
Mentor: Dr. Yagya Raj Pandeya

Kathmandu University
Department of Artificial Intelligence

March 4, 2026

Original Contribution (Lv et al., 2023):

- Identified a fundamental flaw in standard Softmax: training goal ($p_t \rightarrow 1$) mismatches test goal ($p_t > p_j$)
- Proposed **Adaptive Sparse Softmax (AS-Softmax)**:

$$p_t - p_{i \neq t} \geq \delta, \quad \delta \in (0, 1]$$

- Key insight: Easy samples are automatically discarded when condition is satisfied
- Empirically: Better performance, faster training, loss correlates with accuracy ($\rho \approx -0.95$)

But: The margin δ is a **global scalar** — same for all class pairs, all samples, all training time.

Research Gap: Three Unanswered Questions

Question 1: Are all class pairs equally difficult?

In sentiment analysis: distinguishing “very negative” vs. “negative” is harder than “very negative” vs. “very positive.” Should they have the same margin δ ?

Question 2: Are all samples equally ambiguous?

A clearly positive review vs. a neutral one — should the model be equally confident for both?

Question 3: Should margins be static throughout training?

Early training needs to learn basic patterns; later training should focus on hard cases. Should δ adapt over time?

The fixed margin δ is a mathematical simplification, not an optimal solution.

Proposed Research: Generalized Adaptive Margins

Core Idea: Replace the scalar δ with a more expressive mathematical object.

Three Directions of Generalization:

- 1 **Class-Dependent Margins:** $\delta_{t,j}$ — margin between target class t and non-target class j
- 2 **Sample-Dependent Margins:** $\delta(x)$ — margin depends on input sample x
- 3 **Time-Adaptive Margins:** $\delta(t)$ — margin evolves during training

General Form:

$$p_t - p_j \geq \delta_{t,j}(x, t) \quad \forall j \neq t$$

Direction 1: Class-Dependent Margins $\delta_{t,j}$

Motivation: Some class pairs are inherently harder to distinguish.

Mathematical Formulation:

$$p_t - p_j \geq \delta_{t,j}, \quad \delta_{t,j} \in (0, 1]$$

How to Define $\delta_{t,j}$:

- **Semantic Similarity:** $\delta_{t,j} = \delta_{\min} + (\delta_{\max} - \delta_{\min}) \cdot \text{sim}(e_t, e_j)$
(using BERT/Word2Vec embeddings)
- **Confusion-Based:** $\delta_{t,j} \propto \text{Confusion}(t, j)$ from a pretrained model
- **Learnable Low-Rank Factorization:** $\delta_{t,j} = \sigma(u_t^T v_j)$
where $u_t, v_j \in \mathbb{R}^k$, $k \ll n$ (parameter efficient)

Key Mathematical Insight: The margin matrix $M \in \mathbb{R}^{n \times n}$ likely has low-rank structure — classes form clusters with similar inter-cluster margins.

Direction 2: Sample-Dependent Margins $\delta(x)$

Motivation: Not all samples of the same class are equally clear.

Mathematical Formulation:

$$p_t - p_j \geq \delta(x) \quad \forall j \neq t$$

where $\delta(x)$ is a function of the input x .

Implementation Approaches:

- **Confidence-Based:** $\delta(x) = \sigma(\text{entropy}(p(x)))$
Higher uncertainty $\rightarrow$ smaller margin
- **Learnable:** Small neural network $h_\phi(x) \rightarrow \delta(x)$
Train jointly with classifier
- **Feature-Norm Based:** $\delta(x) \propto \|f(x)\|$ where $f(x)$ is feature vector
(inspired by SphereFace, CosFace)

Key Question: Can we prove that sample-dependent margins lead to better-calibrated probabilities?

Direction 3: Time-Adaptive Margins $\delta(t)$

Motivation: Training dynamics change over time.

Mathematical Formulation:

$$p_t - p_j \geq \delta(t) \quad \forall j \neq t$$

Possible Schedules:

- **Increasing:** $\delta(t) = \delta_{\min} + (\delta_{\max} - \delta_{\min}) \frac{t}{T}$
(start easy, get harder)
- **Decreasing:** $\delta(t) = \delta_{\max} - (\delta_{\max} - \delta_{\min}) \frac{t}{T}$
(start strict, relax — similar to AS-Softmax warm-up)
- **Cyclical:** $\delta(t) = \delta_{\text{base}} + \delta_{\text{amp}} \sin(2\pi t / T_{\text{cycle}})$
(explore different regimes)

Key Question: What is the *optimal* schedule? Can we derive it from gradient flow dynamics?

Research Questions

RQ1: Class-Dependent Margins

Can class-pair specific margins $\delta_{t,j}$ improve classification performance, especially on datasets with hierarchical or semantic structure?

RQ2: Sample-Dependent Margins

Does making margins depend on input ambiguity lead to better-calibrated probability estimates?

RQ3: Time-Adaptive Margins

What is the optimal annealing schedule for margins, and can it be derived from first principles?

RQ4: Theoretical Foundations

Can we prove generalization bounds that depend on the margin matrix, showing that adaptive margins lead to tighter bounds?

Research Gap: Why This is Novel

Approach	Class-Pair	Sample-Dep.	Time-Adapt.
AS-Softmax (2023)	No	No	No (warm-up only)
AM-Softmax (2018)	No	No	No
X2-Softmax (2024)	Angular space only	No	No
Largest Margins (2022)	Class-global only	No	No
Our Proposal	Yes	Yes	Yes

Key Novelty:

- First to propose **class-pair margins in probability space**
- First to explore **sample-dependent margins** in softmax variants
- First systematic study of **optimal margin schedules**

Experimental Plan

Datasets (following AS-Softmax):

- SST5 (sentiment, 5 classes, hierarchical structure)
- Clinc_os (intent, 151 classes, semantic relationships)
- Conll2003 (NER, 9 classes, clear class types)
- SIGHAN2015 (Chinese spelling, 5201 classes, large output space)

Baselines:

- Standard Softmax
- AS-Softmax (fixed δ)
- X2-Softmax (adapted to probability space)
- Our proposed variants

Metrics:

- Accuracy / F1 score
- Expected Calibration Error (ECE)
- Training time / convergence speed
- Masked sample ratio per class

Expected Contributions

- 1 **Mathematical:** Generalization of AS-Softmax to class-pair, sample-dependent, and time-adaptive margins; theoretical analysis of margin properties (low-rank structure, metric properties)
- 2 **Empirical:** Demonstrated improvements on text classification benchmarks, especially for datasets with hierarchical class structure
- 3 **Practical:** Parameter-efficient implementation via low-rank factorization; guidelines for margin scheduling
- 4 **Theoretical:** Generalization bounds that depend on margin matrix structure

Timeline

Phase	Activities
Weeks 1-2	Reproduce AS-Softmax results; literature review (X2-Softmax, Largest Margins)
Weeks 3-4	Implement class-dependent margins (semantic similarity version)
Weeks 5-6	Experiments on SST5, Clinc_os; analyze results
Weeks 7-8	Implement learnable low-rank margins
Weeks 9-10	Explore sample-dependent and time-adaptive variants
Weeks 11-12	Theoretical analysis; write first draft

Deliverables:

- Code repository (PyTorch implementation)
- Technical report / conference paper (ACL, EMNLP, or ICLR)
- Mathematical appendix with proofs

Open Questions for Discussion

- **Priority:** Which direction should we explore first? (Class-dependent seems most straightforward)
- **Scope:** Should we focus on text classification, or also test on image/audio data?
- **Theory:** What level of mathematical rigor is expected? (Generalization bounds? Convergence proofs?)
- **Collaboration:** Should we reach out to the AS-Softmax authors?
- **Target venue:** Conference (short paper) or journal (full paper)?

References



Lv, Q., Geng, L., Cao, Z., et al. (2023). "Adaptive Sparse Softmax: An Effective and Efficient Softmax Variant for Text Classification." Transactions on Audio, Speech and Language Processing.



To be added after full review. X2-Softmax: Margin Adaptive Loss Function (Expert Systems with Applications, 2024).



Zhou, et al. (2022). "Learning Towards the Largest Margins." International Conference on Learning Representations (ICLR).



Wang, F., et al. (2018). "Additive Margin Softmax for Face Verification." IEEE Signal Processing Letters.